DANIEL KEVIN ADDO

MS, GStat

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RESEARCH PROFILE

Biostatistics Ph.D. candidate with 7+ years of experience leading statistical analyses, collaborating on clinical/academic research, and teaching applied statistical methods. I am skilled in applying novel and robust statistical methodologies, including causal inference, survival analysis, and machine learning, to generate actionable insights across diverse domains. My doctoral research focuses on advancing methods applied in the causal inference domain.

EDUCATION

University of Utah, Salt Lake City, Utah
Ph.D., Biostatistics — May 2026 (Expected)
Advisors: Tom Greene, Ph.D.
Adam Bress, PharmD, M.S.
Dissertation Aims:

- Develop a balance metric that quantifies bias as an implication of missingness in the outcome over time when estimating the average treatment effect.
- Emulate a target trial to estimate the intent-to-treat effect of initiating angiotensin-II receptor blockers (ARBs) versus angiotensin-converting enzyme inhibitors (ACEIs) -based antihypertensive medication regimens on cognitive decline.
- Evaluate alternative approaches for characterizing proteinuria as a mediator of the effects of interventions for chronic kidney disease (CKD) aimed at slowing CKD progression.

Northern Arizona University, Flagstaff, Arizona
M.S. Statistics — May 2021
Advisor: Roy St. Laurent, Ph.D.

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
B.S. Statistics — May 2017
Advisor: Atinuke Adebanji, Ph.D.

PROFESSIONAL EXPERIENCE

University of Utah, Salt Lake City, Utah
Graduate Research Assistant — August 2021 – Present

- Lead statistician on multiple causal inference and survival analysis projects, guiding data analysis and statistical inference.
- Consult with statisticians and clinicians to design and apply methods appropriate for the study estimand.
- Review literature across statistical and clinical domains to inform study designs and methodology.
- Develop and/or critically review Statistical Analysis Plans (SAPs) to ensure clarity and alignment with study objectives.
- Prepare and deliver detailed statistical reports tailored to facilitate cross-disciplinary understanding.
- Develop efficient and reproducible code in R and SAS to support large-scale data analysis and streamline workflows.
- Mentor junior members on the lab team to carry out clinical research using valid statistical analysis.

Northern Arizona University, Flagstaff, Arizona
Applied Statistics Instructor — July 2021 – August 2021

- Taught core statistical concepts and applications during an intensive summer session to a cohort of 10+ students.
- Designed and delivered a structured lesson plan, ensuring clarity of expectations and measurable student progress.
- Adapted teaching strategies to accommodate diverse learning styles and enhance student engagement and comprehension.
- Provided individualized feedback and mentorship to support student success through graded assignments and office hours.

Graduate Teaching Assistant August 2019 – May 2021

- Taught courses in Algebra, Quantitative Reasoning, and Applied Statistics to 300+ students across multiple semesters.
- Awarded Outstanding Graduate Teaching Assistant at the master's level for excellence in instruction and student support.
- Integrated classroom technologies to enhance student engagement, accessibility, and active learning.
- Adapted teaching strategies to transition effectively to remote/hybrid instruction during the COVID-19 pandemic.
- Improved student learning outcomes by iteratively refining teaching methods based on feedback from coordinators and student evaluations.

ZroNet Africa, Accra, Ghana

Data Analyst December 2018 – July 2019

- Prepared and presented actionable data-driven insights to optimize user engagement on the ZroNet app.
- Designed and implemented an infographic scorecard to seamlessly visualize key daily metrics.
- Improved data security and efficiency with a custom-built data leakage monitoring system.

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Teaching Assistant August 2017 – August 2018

- Led weekly tutorials in statistics and probability to reinforce core concepts via problem-solving and discussion.
- Contributed to a peer-reviewed research publication comparing statistical techniques for forecasting in time series.

PEER-REVIEWED PUBLICATIONS

- Andrews, R.M.*, **Addo, D.K.***, Marcum, Z.A., Greene, T.H., Xu, Y., Derington, C.G., Berchie, R., Pajewski, N.M., Cohen, J.B., Pruzin, J. and Supiano, M.A., 2025. Initiation of ARB-vs ACEI-Based Antihypertensive Medication Regimens and Differences in Rates of Cognitive Decline: An Analysis of SPRINT. *Neurology Open Access*, *1*(3), p.e000011.
- Xu, Y., Derington, C.G., **Addo, D.K.**, He, T., Jacobs, J.A., Mohanty, A.F., An, J., Cushman, W.C., Ho, P.M., Bellows, B.K. and Cohen, J.B., 2024. Trends in Initial Antihypertensive Medication Prescribing Among> 2.8 Million Veterans Newly Diagnosed With Hypertension, 2000 to 2019. *Journal of the American Heart Association*, *13*(20), p.e036557.
- Jacobs, J.A., Derington, C.G., Zheutlin, A.R., King, J.B., Cohen, J.B., Bucheit, J., Kronish, I.M., **Addo, D.K.**, Morisky, D.E., Greene, T.H. and Bress, A.P., 2024. Association Between Self-Reported Medication Adherence and Therapeutic Inertia in Hypertension: A Secondary Analysis of SPRINT (Systolic Blood Pressure Intervention Trial). *Journal of the American Heart Association*, *13*(3), p.e031574.
- Jacobs, J.A., **Addo, D.K.**, Zheutlin, A.R., Derington, C.G., Essien, U.R., Navar, A.M., Hernandez, I., Lloyd-Jones, D.M., King, J.B., Rao, S. and Herrick, J.S., 2023. Prevalence of statin use for primary prevention of atherosclerotic cardiovascular disease by race, ethnicity, and 10-year disease risk in the US: National Health and Nutrition Examination Surveys, 2013 to March 2020. *JAMA cardiology*, *8*(5), pp.443-452.
- Zheutlin, A.R., **Addo, D.K.**, Jacobs, J.A., Derington, C.G., Herrick, J.S., Greene, T., Stulberg, E.L., Berlowitz, D.R., Williamson, J.D., Pajewski, N.M. and Supiano, M.A., 2023. Evidence for age bias contributing to therapeutic inertia in blood pressure management: a secondary analysis of SPRINT. *Hypertension*, *80*(7), pp.1484-1493.
- Twumasi-Ankrah, S., Pels, W.A., Nyantakyi, K. and **Addo, D.K.**, 2019. Comparison of Statistical Techniques for Forecasting Malaria Cases in Ghana. J Biostat Biometric App 4 (1): 102 Abstract Background and Aim: The purpose of this study was to determine an appropriate statistical technique for forecasting the monthly Malaria cases in Ghana. *Methods and Materials: Monthly data spanning from January*.

MANUSCRIPTS UNDER REVIEW/REVISION/PREPARATION

- **Addo, D.K.***, Himbert, C.*, Xu, Y., He, T., Derington, C.G., Scharfstein, D.O., Greene, T., Cohen, J.B., Bress, A. and Hardikar, S., 2025. Angiotensin Receptor Blockers (ARBs) Reduce Cancer Risk Compared to Angiotensin-Converting Enzyme Inhibitors (ACEIs): Evidence from a Large Prospective Veterans Cohort. *Under preparation*.
- Derington, C.G., **Addo, D.K.**, Jacobs, J.A., Cushman, W.C., Egan, B.M., Ho, P.M., Raghavan, S., Hess, P.L., King, J., Xu, Y. and Cohen, J., 2025. Real-World Antihypertensive Treatment Escalation After Initial Monotherapy In The Veterans Health Administration. *Under revision*
- Derington, C.G., **Addo, D.K.**, Jacobs, J.A., Xu, Y., Zheutlin, A., Cohen, J. and Bress, A.P., 2025. Antihypertensive Medication Discontinuation Rates By Initial Mono-And Combination Therapy Use In The Veterans Health Administration, 2000-2022. *Under revision*
- Zheutlin, A.R., Chaitoff, A., **Addo, D.K.**, Jacobs, J.A., Derington, C.G., Stulberg, E., Cohen, J. and Bress, A.P., 2025. Antihypertensive Medication Co-Pay And Adherence Among Veterans Receiving Care Within The Veterans Health Administration. *Under review at JAMA IM*
- Sanuade, O.A., **Addo, D.K.**, Smith, J.D., Carroll, A.J., Shimbo, D., Talegawkar, S., Tucker, K.L., Jacobs, J.A., Derington, C.G. and Bress, A.P., 2024. Urinary Sodium and Incident Apparent Treatment-Resistant Hypertension among African American adults: The JHS. *Under review*

PROGRAMMING & TECHNICAL SKILLS

Certification:
- Data Literacy (Datacamp, 2024)
- GStat (ASA, 2024)
- Human Subjects Research (Citi Program, renewed 2024)

Markup Language:
- LaTeX (Advanced)
- Markdown (Advanced)

Statistical & Data Tools:
- RStudio (Advanced)
- SAS (Intermediate)
- Quarto (Intermediate)
- Tableau (Intermediate)
- Python (Beginner)

AWARDS AND HONORS

Population Health Sciences Travel Award Apr 2024
Department of Population Health Sciences, University of Utah

Outstanding Graduate Teaching Assistant, Master's Level Apr 2021
Graduate School, Northern Arizona University, Apr 2021

Johnson-Ratliff Memorial Scholarship Apr 2020
Department of Mathematics and Statistics, Northern Arizona University

VOLUNTEER / LEADERSHIP ACTIVITIES

Member, *Student Advisory Committee, Department of Population Health Sciences, University of Utah.*
Vice-President, *African Students Association, Northern Arizona University.*
President, *The Navigators of Ghana, Kwame Nkrumah University of Science & Technology.*

CONFERENCES / PROFESSIONAL EVENTS

Participant, Joint Statistical Meetings, Portland, OR Aug 2024
Poster Presenter, Society of Epidemiologic Research, Austin, TX Jun 2024
Participant, Utah Data Science Day, Salt Lake City, UT Jan 2024
Participant, Eli Lilly Discovery Day, Indianapolis, IN Mar 2023
Participant, Utah Data Science Day, Salt Lake City, UT Jan 2023